

Anish Kumar

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SUMMARY

Backend & AI Infrastructure Engineer with production experience building **microservice-based, event-driven systems** — **REST APIs**, distributed messaging (**Kafka**, **RabbitMQ**), caching (**Redis**), and containerized deployments on **Kubernetes** across **Azure** and **AWS**. Strong **object-oriented** foundation across **Java**, **Python**, **C++**, with delivery of scalable backend services, **AI/ML inference integration** (LLM, computer vision, document intelligence), and production reliability engineering. Holds a **B.Tech in Computer Science** from **BML Munjal University**.

EDUCATION

B.Tech, Computer Science & Engineering
BML Munjal University, Gurugram

CGPA: 8.08 / 10
2020 – 2024

EXPERIENCE

Centific — Associate Application Engineer
Chennai, India

Oct 2024 – Aug 2026
Full-Time

Backend Engineering — Loop 1.0

- Designed and owned **microservice-based backend APIs** and **PostgreSQL**-backed services for Loop 1.0, an annotation platform later acquired by a customer in an **approximately \$1M deal**.
- Delivered customer-requested enhancements across **REST API** behavior, database operations, and service workflows, driving production issue resolution for a live, high-usage platform.
- Applied **clean-code** and **design-pattern** principles to keep services maintainable and extensible across multiple customer use cases.

Event-Driven & Distributed Systems

- Built **event-driven microservices** with **Kafka**, **RabbitMQ**, **Redis**, and **PostgreSQL** to decouple ingestion, asynchronous processing, caching, and operational storage.
- Implemented **Redis caching** strategies to reduce latency and offload primary datastores under production traffic.
- Containerized backend and data services with **Docker** and deployed on **Kubernetes**, improving release consistency and reducing manual operational effort.
- Operated production systems with **Prometheus**, **Grafana**, structured logging, database analysis, and queue inspection to improve reliability and incident response.

AI/ML Integration & Model Serving

- Integrated AI-driven capabilities into backend services — **LLM inference (Llama 70B)**, computer vision (**YOLOv8/v11**, **NVIDIA VILA & DINO NIM**), and document intelligence (**PaddleOCR**).
- Deployed **GPU-backed model-serving** workflows using **Hugging Face**, **NVIDIA NGC**, **Azure**, and **RunPod** for scalable inference delivery.
- Secured service configuration and secrets with **JWT**, **OAuth**, **Azure Key Vault**, and **HashiCorp Vault**.

Movidu Tech. — Data Analyst Intern
Bengaluru, India

Feb 2024 – Aug 2024

- Built **Python ETL** workflows for large datasets — extraction, validation, transformation, and structured storage for analytics.
- Conducted **exploratory data analysis and preprocessing** on raw datasets in **Jupyter notebooks** using **Pandas**, **NumPy**, and **Scikit-learn**, preparing and validating data ahead of drought prediction modeling.
- Developed logistics dashboards and optimized **route-planning logic**, surfacing operational exceptions for data-driven decisions.

Adoptev — Web Development Intern
Gurugram, India

Jun 2022 – Jul 2022

- Designed complete **wireframes and UI mockups in Figma** for the website's information architecture prior to development.
- Built responsive **ReactJS + Tailwind** components and debugged API integrations to improve cross-device workflow reliability.

PROJECTS

Multi-Workspace AI Document Assistant — RAG & Tool Calling

[GitHub](#)

FastAPI, React, TypeScript, PostgreSQL + pgvector, Gemini, Groq, Docker, GitHub Actions, Render, Neon

- Built and deployed an end-to-end **AI data assistant**: users chat with their documents in isolated workspaces, with **multi-tenant RAG** on one shared pgvector table (isolation enforced inside the SQL vector query), **hybrid vector + full-text search (RRF)**, and server-validated citations with calibrated “I don’t know” refusals.
- Designed an **LLM tool-calling agent loop** (task creation, Discord notifications, multi-step search) with **schema-validated arguments**, per-turn limits, audit logging, and **prompt-injection resistance**; **SSE token streaming** with Gemini → Groq model fallback.
- Shipped argon2 + httpOnly-JWT auth, idempotent ingestion, an observability dashboard (latency, tokens, retrieval hit rate), and **80+ tests** in CI that boots the production Docker image.

ATS Optimize — AI Resume Matchmaking Platform

[GitHub](#)

FastAPI, React, PostgreSQL, JWT, OAuth, Vite, Render, Vercel

- Built a full-stack **matchmaking platform** that parses resumes and scores them against job descriptions using custom NLP — a **recommendation/ranking engine** producing keyword-match matrices and compatibility breakdowns.
- Engineered a deterministic AI rewriting module (proper-noun restoration, weakness detection) to improve output **without hallucination**.
- Implemented full **auth** (JWT access/refresh in httpOnly cookies, Google OAuth, email verification) with cross-domain cookie, CORS, and environment-based deployment handling.

Distributed Task Processing Platform

[GitHub](#)

Python, FastAPI, Apache Kafka, PostgreSQL, Redis, Docker, SQLAlchemy, Prometheus, Grafana, pytest, GitHub Actions

- Built a 9-service, **event-driven** task processing platform using **Apache Kafka**, **PostgreSQL**, and **Redis**, scaling horizontally through Kafka **consumer groups** with `task_id` partitioning for per-task ordering and cross-task parallelism.
- Implemented **Transactional Outbox** for atomic task-state and event persistence with **at-least-once delivery and idempotent execution**; engineered fault tolerance via **exponential backoff with jitter**, a non-blocking retry topic, a **dead-letter queue**, and lease-based recovery for crashed workers.
- Instrumented **26 Prometheus metrics** and a **23-panel Grafana** dashboard; validated by **210 automated tests** (94 integration tests running against live Kafka/PostgreSQL/Redis) in a 3-stage **GitHub Actions CI** pipeline with end-to-end smoke tests.

TECHNICAL SKILLS

Languages: Java, Python, SQL, C++

Backend & Frontend: Microservices, REST API design, FastAPI, Flask, SSE streaming, React, TypeScript

Event-Driven & Async: Kafka, RabbitMQ, Redis, event-driven architecture

Databases & Caching: PostgreSQL, pgvector, SQL (incl. MySQL), Redis, Qdrant

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Azure DevOps, Git, CI/CD, JIRA, Confluence

AI / ML Systems: RAG, LLM tool/function calling, hybrid search, LLM inference, semantic search, PyTorch, TensorFlow, Hugging Face, NVIDIA NIM

Observability: Prometheus, Grafana, logging, monitoring

Security: JWT, OAuth, Azure Key Vault, HashiCorp Vault

ACHIEVEMENTS

- Student Councilor – 2023
- Sponsorship Lead – 67th Milestone College Festival